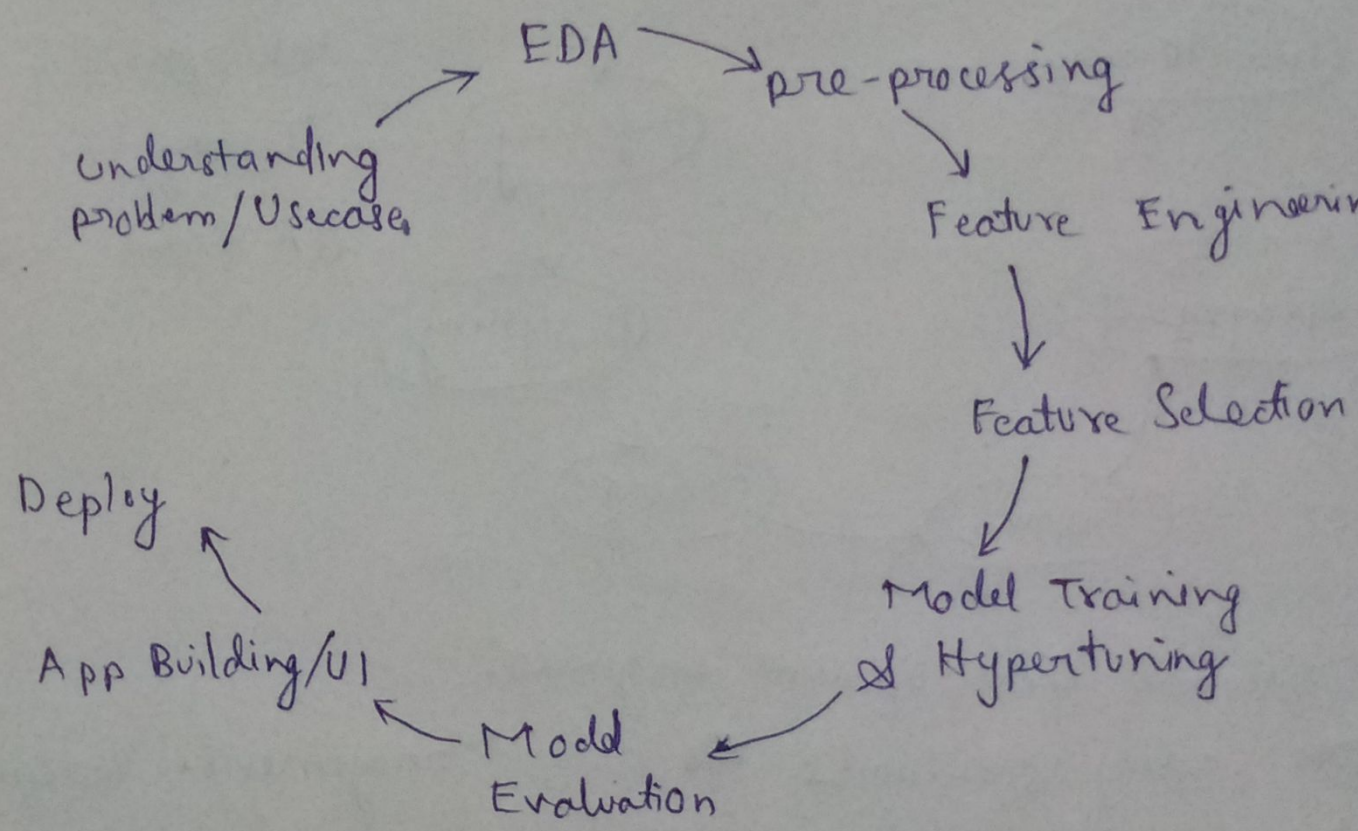
Date - 28. Apr 2026

MLOps Introduction

MLOps

- Machine learning operations (MLOps) are a set of practices that automate and simplify machine learning (ML) workflows and deployments.
- it is the development and use of machine learning models by development operations (DevOps) teams.
- MLOps adds discipline to the development and deployment of machine learning models, making the development process more reliable and productive.

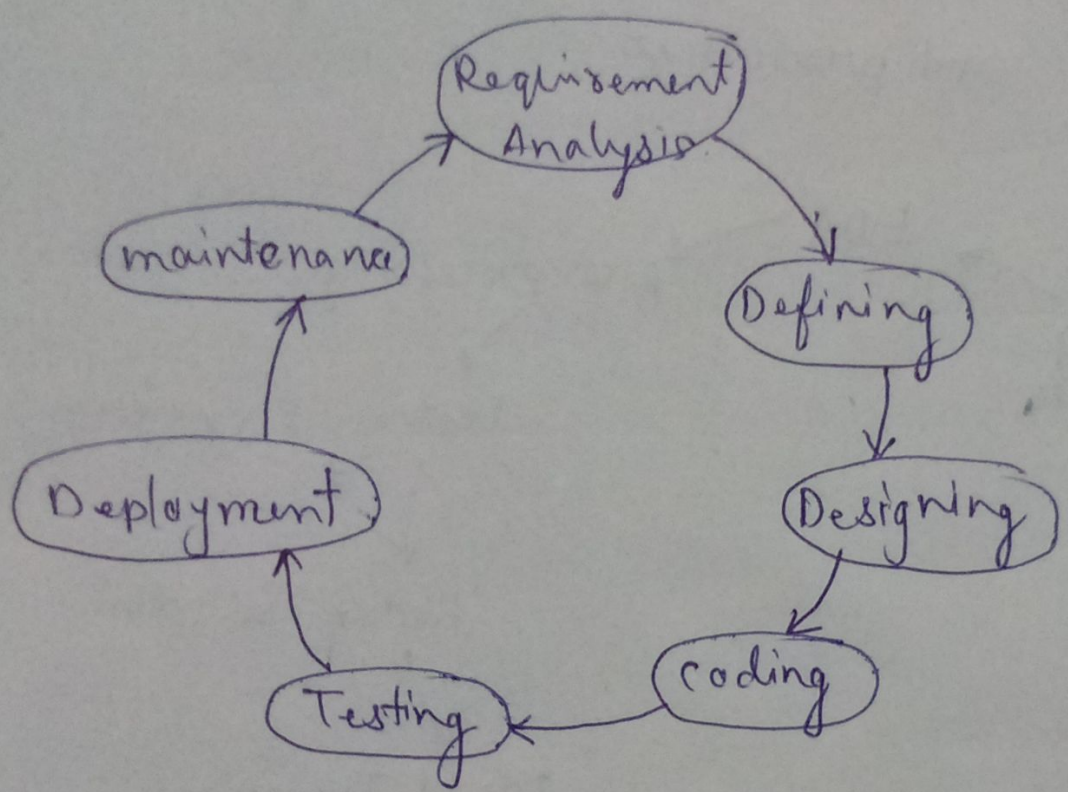


Issues with DS practice without MLOps

1. Low Coding standards
   OOPs concept, Modular Coding, logging, exception
2. No Data Management
   Data Ingestion/Artifacts
3. Versioning
   Code, Data, Model

4. Datapipeline / Experiments
5. No CI/CD concept
6. Scalability & Monitoring (Production)
   Kubernetes, Prometheus, Grafana
7. Cross team Friction.

Software Development life cycle (SDLC)



Dev	Ops
Development	Delivery
Testing	Deployment
QA Team	Maintenance

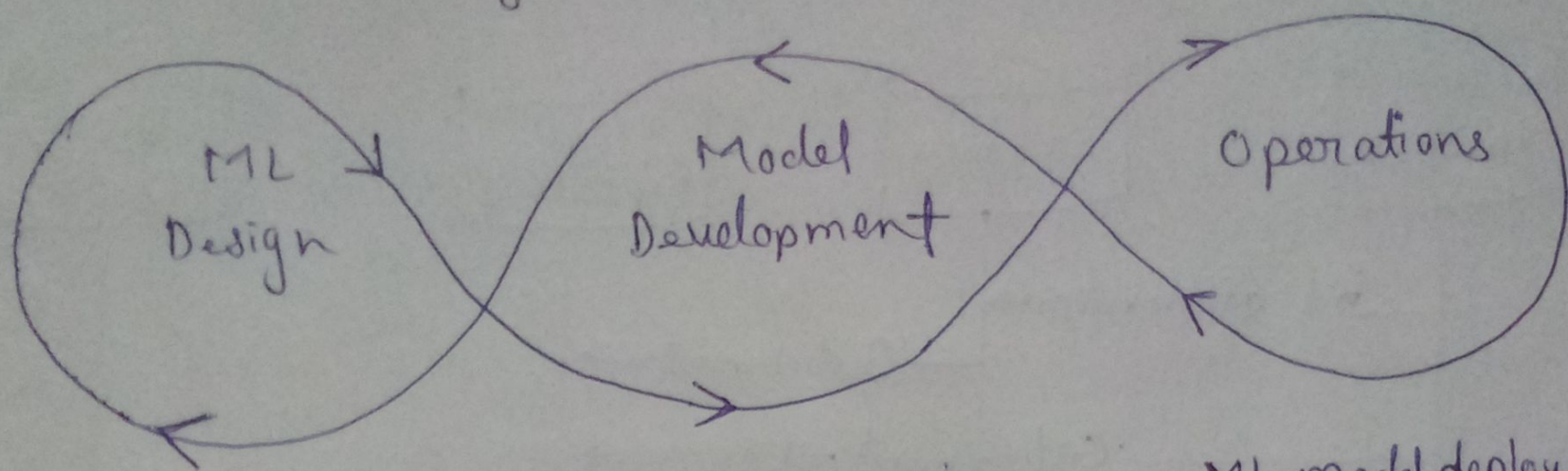
1. Software Development engineer:
   coding standards, data Management, Versoning, Data pipeline/Experiments

2. Operations:
   CI/CD concept, Scalability and Monitoring, cross team friction.

SDE + Operations ⟶ MLOps

Data Engineer + Data Scientist + SDE + Operation ⟶ MLOps Engineer

Machine learning Operations (MLOps)



Gather requirements
prioritize ML us cases
Business Understanding
Data Acquisition

Data pre-processing
Feature Engineering
Model training/Experiment
Model analysis evaluation

ML model deployment
CI/CD pipelines
model monitoring and Triggering